

# SSV Conclusions B

A concrete superfluid holographic screen undershoots Newton’s  
constant by 38 orders: a mode-counting negative

Stig Norland

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## Abstract

This note records the sharpest quantitative result the Saturated Superfluid Vacuum (SSV) programme produced about emergent gravity, and it is a negative. Entropic and holographic gravity (Jacobson, Verlinde, Padmanabhan) keep the holographic screen *abstract*. SSV’s distinctive move was to give the screen a concrete microscopic substrate — a logarithmic-Schrödinger condensate whose self-defined acoustic horizon carries an area-law entanglement entropy that can be *computed* rather than assumed. The form survives: the entanglement entropy is area-law, as expected for a local quadratic Hamiltonian. The magnitude does not. Counting the actual degrees of freedom the substrate offers per grain gives  $\mathcal{O}(10)$  Bogoliubov modes — a few bits — against the  $N \simeq 1.7 \times 10^{38}$  required to reproduce  $G$ . The screen, made literal and counted, misses Newton’s constant by roughly  $10^{37}$ . This is the Sakharov / Susskind–Uglum species problem, here *demonstrated by direct mode-counting* rather than argued, and it is a clean constraint on any reading that takes a holographic screen to be a real reservoir of local degrees of freedom. Per the repository’s rule, the negative is stated at full strength.

## 1 The setup: making the screen literal

In the thermodynamic route to gravity, the field equation follows from  $\delta Q = T dS$  applied to local causal horizons (Jacobson 1995), with the entropy taken to be area-law,  $S = A/4\ell_P^2$ . Verlinde’s entropic-force reading and Padmanabhan’s surface-thermodynamics share this skeleton. In all of them the holographic screen is a bookkeeping surface: one assumes an area-law entropy with the Planck-scale coefficient and does not ask what physical degrees of freedom supply it.

SSV’s one genuinely distinctive contribution is to refuse that abstraction. The vacuum is a real medium with order parameter  $\Psi = \sqrt{\rho} e^{i\theta}$  governed by a logarithmic Schrödinger equation; the screen is a self-defined acoustic horizon, the surface where the background flow turns sonic (Visser). Because the medium is concrete, two things that are normally assumed become *computable*: the area law, and the coefficient.

## 2 The form survives (and is the cheap half)

Quantising the LogSE Bogoliubov fluctuations and computing the vacuum entanglement entropy by the correlator method (Peschel; Casini–Huerta) via the Srednicki radial decomposition gives an area law. Across a static acoustic inhomogeneity (a “dumb hole”) the scaling exponent tracks a provably area-law gapped-scalar calibrator and sits a full  $\sim 1.2$  in log–log slope below a volume-law thermal control; the entropy-per-area is universal at the few-percent level; and the chiral-shear term is *exactly* silent (`instruments/paper_v/horizon_entanglement_entropy.py`).

This must be stated honestly (repository rule 1): the area law is the *expected* side. It follows from the LogSE quadratic Hamiltonian being *local* — generic for any local free field (Srednicki 1993; Bombelli–Koul–Lee–Sorkin 1986; Jacobson’s entanglement-equilibrium, 2016). So the computation *earns* (turns assumed into derived-from-locality) the area law that the SSV gravity papers had merely asserted by cell-counting, but it is not a surprise and not a discovery. The discriminator was shown to be negative-capable — it correctly rejects volume-law data — which is what makes the next step trustworthy.

### 3 The magnitude fails (and is the whole point)

The coefficient is the test. Writing  $\delta Q = T_H dS$  with the computed area-law  $dS = \eta dA$  closes the Jacobson construction to  $\nabla^2 \Phi = 4\pi G_{\text{eff}} e/c^2$  with  $G_{\text{eff}} = 1/(4\hbar c \eta)$  (`instruments/paper_v/dumb_hole_surface`). To recover the measured  $G$ , the entropy density  $\eta$  must correspond to

$$N = \frac{1}{\alpha_G} = \frac{\hbar c}{G m_p^2} = \left( \frac{m_P}{m_p} \right)^2 \simeq 1.7 \times 10^{38} \quad (1)$$

microstates per grain — this is just the Planck-to-proton hierarchy.

Now count what the substrate actually provides. The localised excitations of a LogSE vortex core — the bound Bogoliubov (BdG) modes — number  $\mathcal{O}(10)$ , i.e. a few bits, per healing-length cell (`instruments/paper_i/vortex_core_mode_spectrum.py`). The continuum modes are delocalised and do not localise entropy on the grain. So the concrete screen supplies of order  $10^1$  states where  $10^{38}$  are needed: a shortfall of about  $10^{37}$ , and correspondingly  $G_{\text{eff}}$  overshoots the Newtonian value by  $(\bar{\lambda}_p/\ell_P)^2 \approx 1.7 \times 10^{38}$ .

### 4 What the negative establishes

When a holographic screen is given a real microscopic substrate and its localised degrees of freedom are *counted*, the area law emerges for free but the Planck-scale coefficient does not: the count is short by  $\sim 38$  orders of magnitude. A holographic screen that is a literal reservoir of local degrees of freedom cannot, in this construction, reproduce  $G$ . Newton’s constant must enter below the effective theory’s cutoff — as a trans-Planckian input, not an emergent output.

This is the Sakharov induced-gravity problem and the Susskind–Uglum species problem, which Volovik identifies as the structural weak point of superfluid gravity. The contribution here is not the existence of the gap but its *concrete demonstration*: rather than estimating an induced Newton constant from a cutoff, one builds the medium, computes its horizon entanglement entropy, counts its grain degrees of freedom, and reads off the shortfall directly. That makes the failure model-specific and checkable rather than a heuristic worry.

### 5 Caveats, stated plainly

- The identity  $(\bar{\lambda}_p/\ell_P)^2 = 1/\alpha_G$  is a *tautology*: it restates the hierarchy and is not a derivation. It must not be read backwards as “deriving  $N$ .” The negative is the *measured* count ( $\mathcal{O}(10)$ ), confronted with this target.

- Bound modes are finite by construction; continuum modes are delocalised. A genuinely different substrate (e.g. a rank-2,  $^3\text{He-A}$ -like order parameter with sub-grain structure) is not excluded by this argument — it simply is not the LogSE medium, and would have to supply its own  $10^{38}$  states.
- The area-law computation assumes irrotational flow for the exact chiral silence; a rotational/vortex-sink horizon carries a bounded residual risk that was not closed.

None of these weaken the headline: in the medium actually proposed, the screen is  $\sim 37$  orders too sparse to be the source of  $G$ .

## References

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